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Alginate-Based Edible Coating to Preserve the Quality and Extend the Shelf Life of Fresh-Cut Salad

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Abstract

The food industry is actively seeking solutions to reduce or replace conventional petroleum-based plastic packaging and, at the same time, to identify strategies that limit the rapid deterioration of fresh products. In this context, the present study evaluated the effectiveness of an edible emulsion coating based on lemongrass essential oil and alginate in delaying the spoilage of *Lactuca sativa* salad. Following rheological investigation, 1% alginate emulsion was selected as the coating formulation and applied by spraying onto fresh-cut lettuce, and the effect of the treatment was monitored throughout storage. Fresh-cut *Lactuca sativa* salad was assessed in terms of weight loss, pH, titratable acidity, visual appearance, sensory analysis, and microbiological contamination. Measurements of weight loss, pH, and titratable acidity indicated the lack of significant differences between coated and uncoated salads leaves. However, coated samples exhibited improved quality in the first 8 days of storage, particularly with evidence of a reduction in psychrotrophic and mesophilic bacteria. The proposed coating also helped to preserve the visual appearance of the leaves, with no visible browning during storage, and the sensory evaluation results were encouraging. Overall, these findings suggest that the technology investigated is promising for supporting the use of emulsion-based edible coatings to reduce the rapid spoilage of *Lactuca sativa* salad during storage.

Keywords: edible coating; sodium alginate; *Lactuca sativa*; emulsion; rheology; spoilage rate

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1. Introduction

In recent years, consumer demand for ready-to-eat foods, including products based on fresh fruits and vegetables, has steadily increased, becoming a significant component of modern diets. Such products are highly appreciated for their convenience and for nutritional aspects, as they represent a valuable source of vitamins, minerals, and fibres.

Fresh-cut salad leaves are an example of widespread diffused ready-to-eat products. This kind of convenient product is typically packaged in 100–200 g single-use plastic bags made from petroleum. These packaging solutions pose a serious environmental issue, contrasting with the need to reduce the use of non-recyclable materials and minimize food waste [1].

The food industry is, therefore, increasingly oriented toward the identification of innovative, eco-friendly alternatives for a more rational use of packaging [2]. Recently, the European Union has introduced a new regulation [3] as part of a strategy to reduce the use of plastic packaging from 2026 onwards, which may lead to a ban on packaging for fresh fruit and vegetables weighing under 1.5 kg, including bagged salad.

Among the various leafy vegetables used in ready-to-eat fresh-cut salads, *Lactuca sativa* (lettuce) is the most widely consumed. To obtain fresh-cut lettuce, minimal processing operations (selection, cutting, washing, rinsing, drying) are generally carried out using whole heads followed by packaging in air or modified atmosphere and refrigeration [4,5].

During storage, packaged lettuce undergoes a series of physicochemical changes, including colour change and loss of sensory quality, influenced by various factors, such as maturity at harvest, processing conditions, final packaging environment, and microbial load [6]. The latter should not exceed 6–7 log CFU/g (mesophilic microorganisms), and the product must be free from human pathogens [7].

Appropriate post-harvest management is, therefore, essential to maintain overall quality and extend the shelf life of fresh-cut produce, with positive effects throughout the production chain.

Edible coatings have emerged as a promising strategy to increase the shelf life of fresh fruits and vegetables by limiting microbial growth and slowing enzymatic browning [8]. Their main components are hydrophilic food-grade biopolymers (hydrocolloids), which when dispersed in water, form gel-like suspensions. These hydrocolloids, including selected carbohydrates and proteins, possess film-forming properties [9].

Among biopolymers, sodium alginate—a polysaccharide derived from bacteria and brown algae—is a versatile, non-toxic, and inexpensive material with interesting technological properties. From a rheological point of view, sodium alginate exhibits non-Newtonian shear-thinning behaviour, which facilitates its use as a coating. In addition, its viscoelastic properties enable uniform coverage and adaptability to irregular surfaces [10,11].

To date, only a few studies have examined the use of edible alginate coatings on fresh-cut lettuce. Tay and Perera [12] reported that the crispiness of fresh-cut leaves of baby butterhead lettuce coated with alginate crosslinked with calcium ions was preserved for more than 10 days. In 2021, Li et al. [13] observed that lettuce treated with an alginate coating showed an improved appearance, slowed browning process, and delayed senescence. In the same year, Das and coworkers [14] examined the use of an alginate- and vanillin-based film as packaging for ready-to-eat salads.

When hydrocolloids are combined with oil fractions, such as essential oils (EOs), they give rise to dispersions in which the mechanical properties of hydrocolloid matrix are complemented by the antimicrobial activity of EOs. Several studies have discussed the use of EO to produce emulsion-based coatings, as these natural compounds are generally well-accepted by consumers and researchers, mainly for their antimicrobial and antioxidant properties [15]. Incorporation of EOs into emulsions allows the use of low concentrations of these compounds, thereby limiting the negative effects on sensory quality while maintaining product safety [16]. For example, an emulsion containing only 0.1% oregano oil was found to be effective against foodborne pathogens such as *L. monocytogenes*, *S. typhimurium*, *S. aureus*, and *E. coli* in lettuce [17]. In a recent study, Xylia et al. [18] evaluated the effectiveness of oregano, ascorbic acid chitosan, and their combinations on quality attributes of fresh-cut lettuce and reported that the combination of oregano EO and chitosan reduced yeast and mould counts after four days of application. Other studies have also demonstrated that the efficacy of emulsions based on hydrocolloids and essential oil in prolonging the shelf life of fresh-cut rocket salad was due to the antimicrobial activity of the edible coating [15].

In this study, *Cymbopogon nardus* EO was selected, because previous studies have shown that when incorporated at low concentrations into alginate-based dispersions, its strong aroma is markedly attenuated while its antimicrobial and antioxidant activities are preserved. These properties have been shown to help maintain the sensory and visual quality of leafy vegetables, including lettuce [16,19].

Earlier studies have reported that alginate–EO edible coatings are effective in preserving the quality and nutritional properties of various fresh products, including bananas (*Musaceae* family), Salanova lettuce, and apple slices [19–21]. In particular, its application on ready-to-eat Salanova lettuce protected bioactive pigments and phenolic compounds, reduced moisture loss, and maintained visual quality.

Despite the numerous applications of alginate and essential oil-based coatings on various fruit and vegetable matrices, the literature reports only a few targeted studies on fresh-cut lettuce, and these are often limited to individual quality parameters. Furthermore, joint assessments of chemical, microbiological, and sensory parameters under realistic storage conditions for the fresh-cut industry are lacking.

This study aims to fill these gaps by proposing an edible alginate–essential oil emulsion as a post-harvest technology for fresh-cut *Lactuca sativa*, evaluated in an integrated manner and within a rapidly evolving European regulatory context.

2. Materials and Methods

2.1. Materials, Reagents, and Standards

Iceberg lettuce (*Lactuca sativa* L.) was purchased from a local retailer (Campobasso, Italy) and stored at 4 °C until use (within 24 h). *Cymbopogon nardus* essential oil (Erbamea, San Giustino, Perugia, Italy) was obtained from a local herbal store. The product was a 100% pure, food-grade essential oil, commercially available for dietary use. Food-grade sodium alginate was obtained from Farmalabor (Canosa Di Puglia, Italy), and all the other chemical reagents were analytical grade and purchased from Merck (Merck, Darmstadt, Germany).

2.2. Edible Coating Preparation

Emulsions were prepared following the previously described procedure [20]. In summary, suspension of sodium alginate (1 and 2% w/w) was prepared by dissolving the polymer in ultrapure water at 70 °C under constant stirring for approximately 2 h. The stock emulsion was obtained by homogenizing the alginate suspension in the presence of Tween 80 5% w/w and essential oil 0.5% w/w using a laboratory T25 digital Ultra-Turrax mixer (IKA, Staufen, Germany) with an S25N-8G probe, working at 24,000 rpm for 4 min. After the homogenization the samples were subjected to ultrasonication (Ultrasonic Homogenizer Model 300 VT, BioLogics Inc., Manassas, VA, USA) for 1 min at 120 W with 50% pulsed frequency (1 s on / 1 s off). The stock emulsion was then diluted with the same continuous phase (alginate 1 or 2% w/w), mixed through homogenization for 2 min with an S25N-18G probe in order to obtain emulsions with final EO concentration 0.1% (w/w) and final Tween 80 concentration 1% (w/w).

2.3. Rheological Characterization

Rheological characterization of edible coating suspensions was made through a rotational rheometer, Haake MARS III (Thermo Scientific, Karlsruhe, Germany). A 60 mm-diameter parallel plate (PP60) probe was used with a gap distance of 1 mm. Flow curves were collected at 25 °C, and the temperature was controlled by a Peltier element in combination with a water bath system (Phoenix II, Thermo Scientific, Karlsruhe, Germany).

2.4. Preparation and Coating for Salad Samples

Salad leaves were washed by immersion in tap water for several minutes, drained, and cut with a stainless-steel knife. Excess moisture was removed by manual centrifugation, followed by blotting with absorbent paper. Emulsion was applied to the lettuce leaves using a laboratory airbrush (Model 350-9 Badger Ari-Brush Co., Bellwood, IL, USA). Salad samples coated with the proposed emulsion were referred to as EC, while the uncoated samples (as reference control) were referred to as UC. All salad portions (coated and uncoated) were placed on perforated plastic trays for drying in a forced air dryer for approximately two minutes. Once dried, samples of about 10 g, accurately weighed, were placed in polyethylene terephthalate (PET) plastic bowls with a capacity of 150 cm^3, sealed with a lid, and finally stored in a refrigerator at 4 °C. For the microbiological analysis, after drying, coated (EC) and uncoated (UC) pieces of salad were taken separately and placed in plastic zip-lock bags and stored at 4 °C.

2.5. Physicochemical Analysis

Weight loss was determined in percentage by comparing the initial weight of the complete plastic bowl containing the salad sample with its weight measured at designated storage intervals. pH and titratable acidity (expressed as % citric acid (w/w)) were determined on the salad leaf aqueous extract. The aqueous extract was obtained by homogenizing 5 g of leaf fragments with distilled water; the homogenate was filtered and brought to a final volume of 50 mL with water. pH was measured through a pH metre and titratable acidity was obtained by titrating a specific volume of aqueous extract with a 0.1 M NaOH solution until reaching a pH of 8.2.

Hunter's colour values (L*, a*, b*) were measured on EC and UC samples at time 0, after the treatment, and after 8 and 14 days of storage using a Minolta CR-300 chromameter (Konica Minolta, Japan). The instrument was calibrated with a standard white plate before sampling lettuce leaves. The obtained data were used to calculate the total colour change, applying the following equation: $\Delta E\ [(\Delta L^2 + \Delta a^2 + \Delta b^2)^{1/2}]$. The differences ($\Delta E$) were with reference to the initial L*, a*, and b* values at 0 days of storage of each sample. At least six different measurements were performed [22].

Qualitative identification of the main components of EO was based on the literature on *Cymbopogon nardus* essential oil [23].

2.6. Sensory Analysis

Sensory evaluation was carried out on coated (EC) and uncoated (UC) salad leaves during storage. The panel consisted of eighteen untrained panellists (men and women, 25–55 years old) from the Department of Agricultural, Environmental, and Food Sciences of the University of Molise (Italy). Analyses were conducted in a sensory laboratory under controlled lighting and odour-free conditions. Samples were stored at 4 °C and equilibrated to room temperature for 15 min prior to evaluation. Each panellist performed two replicates per session. All participants were informed about the nature of the study and provided consent; no personal data were collected.

Both a consumer rating test and a triangle test were performed at 0, 2, 7, and 10 days, within the same sensory session at each storage time; a short pause was allowed between the two evaluations to avoid cross-interference.

For the consumer rating test, samples were presented with random three-digit codes. Before evaluation, panellists received a brief explanation of the 1–5 rating scale to ensure a consistent understanding of the scoring procedure. Panellists then evaluated three sensory attributes, colour, odour, and crispiness (assessed by touch), and rated their perceived intensity on a five-point scale (1 = absence of perception; 5 = maximum intensity). As the assessors were untrained, these scores reflect consumer perception rather than calibrated descriptive analysis.

For the triangle test, performed according to the UNI EN ISO 4120:2021 standard [24], panellists received two trays divided into three sections, each containing three leaves per section. One tray included two sections with UC and one with EC, while the second tray had the opposite arrangement. The position of the different samples within each triad was randomized. Panellists first examined appearance, odour, and texture (by touch), then identified the different samples (forced choice), indicated the attribute responsible for the perceived difference, and expressed their willingness to purchase the identified sample. The significance of the results was determined by comparing the number of correct identifications with the critical values in the unilateral $p = 1/3$ significance table.

2.7. Microbiological Assays

The microbiological quality of fresh-cut lettuce was evaluated by enumerating key spoilage-associated microbial groups: mesophilic and psychrotrophic bacteria, lactic acid bacteria (LAB), yeasts, and moulds. For each sample, 10 g was aseptically transferred into a sterile stomacher bag containing 90 mL of physiological saline solution (0.9% NaCl) and homogenized using a peristaltic mixer (Stomacher BagMixer, Intersciences, France). Serial decimal dilutions were prepared, and aliquots were plated on appropriate selective media. Psychrotrophic and mesophilic bacteria were quantified on Plate Count Agar (PCA), incubated at 20 °C and 30 °C, respectively, for 24–48 h under aerobic conditions. LAB were enumerated on MRS (De Man, Rogosa and Sharpe Agar) supplemented with cycloheximide (0.004 g/L), incubated at 30 °C for 48 h under anaerobic conditions. Yeasts and moulds were assessed on Rose Bengal Agar (RBA), with incubation at 30 °C for 5 days in aerobic conditions. All culture media and reagents were purchased from Merck KGaA (Darmstadt, Germany). The samples were analyzed at time 0, to assess the initial microbial load, and at set time intervals (5, 8, and 14 days). Results are expressed as log colony-forming units (CFUs) per gram of fresh weight.

2.8. Data Analysis

The statistical data analysis was performed using SPSS software (version 23.0, IBM SPSS Statistics, Armonk, NY, USA). Analysis of variance with ANOVA Duncan's post hoc test ($p < 0.05$) allowed the determination of statistically significant dissimilarities between the different samples at the same times and within the same sample at different times. All experiments were performed on three replicates. Means and standard deviations were calculated.

3. Results

3.1. Choice and Characterization of Coating Formulation

A preliminary rheological characterization of the coating formulation at different alginate concentrations (1 and 2%) was explored, with the purpose of selecting the most appropriate type of coating. Salad leaves must be covered by spraying, since other application methods, such as dipping or brushing, would transfer too much material to the leaf surfaces.

Alginate is a biopolymer with colloidal properties suitable for stabilizing emulsions and adhering to surfaces. To this end, two coating formulations containing alginate and EO were developed. The oil content was set at 0.1%, as higher amounts affected the turbidity of the suspension and the overall flavour. Figure 1 shows the flow curves of the formulations containing 1% and 2% alginate. The curves were fitted to the Ostwald de Waale equation (Equation (1)):

$$\tau = k \times \dot{\gamma}^{n}, \tag{1}$$

where τ is the shear stress, $\dot{\gamma}$ is the shear rate, k is the consistency index, and n is the index of rheological behaviour ($n = 1$ for Newtonian fluids, $n < 1$ for shear-thinning fluids, and $n > 1$ for shear thickening fluids).

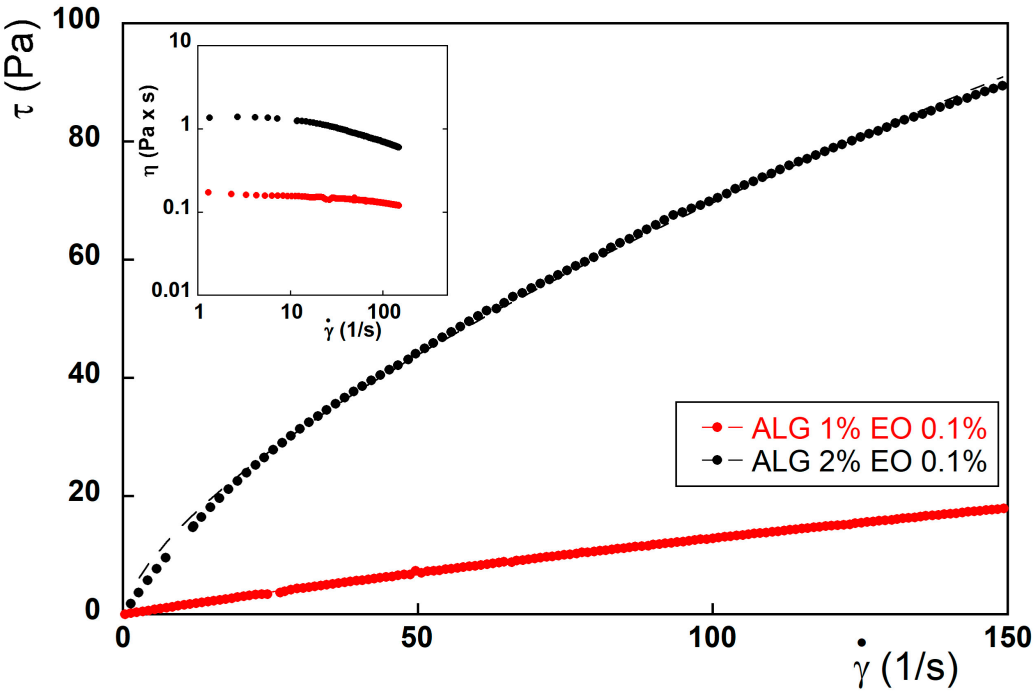

Figure 1. Flow curves of 1 and 2% Alginate suspensions with 0.1% of essential oil. Inset shows the apparent viscosity values as a function of the shear rate.

The flow curves of both suspensions well fitted to the power low equation, and both behaved as shear-thinning fluids. As shown by the parameters reported in Table 1, this shear-thinning character was more pronounced for the 2% alginate suspension, which also showed higher values of viscosity. Based on the presented results, the alginate 1% suspension was selected for the present investigation, as this formulation is expected to be more suitable for spraying applications: excessively high viscosities can cause difficulties with pumps, increased energy consumption, and nozzle clogging.

Table 1. Parameters obtained from flow curves fitting to the Ostwald de Waale equation.

	k	n	R^2
Alginate 1% EO 0.1%	0.237 ± 0.003	0.866 ± 0.003	0.9994
Alginate 2% EO 0.1%	3.266 ± 0.053	0.665 ± 0.004	0.9986

3.2. Physical-Chemical Parameters of Coated (EC) and Uncoated (UC) Salad

Once the alginate 1% suspension was selected as the most suitable for the salad coating, it was applied by spraying onto the salad leaves, and the samples were monitored over a defined storage period. About shelf life, the European Union does not establish a legal regulation for ready-to-eat salads: it is the responsibility of the food business operator to determine the "use-by" date, and to ensure product safety throughout the declared period. According to Arienzo et al. [25], the average shelf life of ready-to-eat fresh vegetables ranges from 5 to 7 days, and, after packages have been opened, products can be stored at refrigeration temperatures lower than 8 °C for a maximum of 2 days. In the present study, the maximum storage period was, therefore, set at 14 days.

Assessing the shelf life of ready-to-eat fruit and vegetables involved monitoring the changes in physicochemical parameters, like weight loss (due to product dehydration, leading to textural changes), pH, and titratable acidity (due to microbial activity and potential spoilage) (Figures 2 and 3).

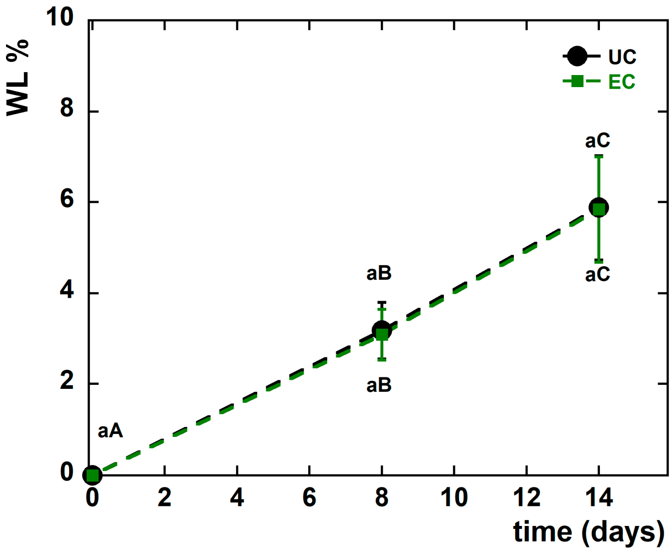

Figure 2. Weight loss (WL, %) changes observed in uncoated (UC—black) and coated (EC—green) salad samples during the storage. Means with the same letters (lowercase: under different treatments for the same time; uppercase: under the same treatment during different storage times) are not significantly different according to Duncan's test ($p > 0.05$). Vertical bars indicate standard deviation.

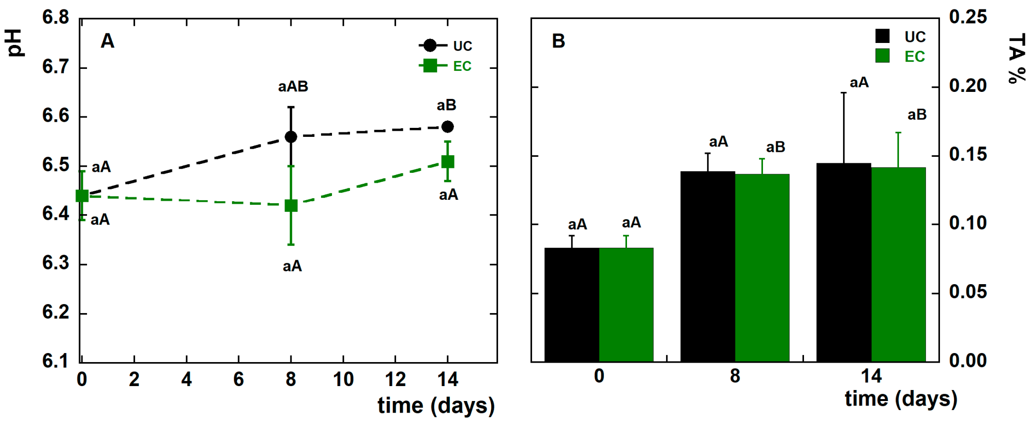

Figure 3. Changes in (**A**) pH and (**B**) titratable acidity (TA, % *w*/*w* as citric acid) of uncoated (UC, black) and coated (EC, green) lettuce samples during storage at 4 °C. Lowercase letters indicate differences under different treatments (UC vs. EC) at the same storage time, whereas uppercase letters indicate differences under the same treatment over time, according to Duncan's test ($p > 0.05$). Vertical bars represent standard deviation.

The weight changes in both coated (EC) and uncoated (UC) lettuce samples are reported in Figure 2.

As shown, all samples exhibited a progressive weight loss over time with no significant differences between the EC and UC samples. Mean weight loss for all samples was approximately 3% and 6%, after 8 and 14 days of storage, respectively. These findings suggest that weight loss was not significantly influenced by the applied coating, likely due to the low storage temperature (4 °C) and the use of plastic bowls closed with lids, conditions that limited water evaporation. Similar behaviour has been observed in previous studies, which described a minimal moisture loss in fresh-cut lettuce stored under similar low temperatures under controlled-atmosphere conditions where lettuce was treated with free or microencapsulated thyme EO [26].

The pH and titratable acidity (TA) values of EC and UC salad leaves, stored at 4 °C and monitored during the storage, are reported in Figure 3A,B, respectively.

Monitoring pH and titratable acidity is particularly relevant in fresh-cut lettuce, as both parameters are sensitive indicators of tissue metabolism and early microbial activity, which in turn influence product stability and shelf life. In our study, although the graphical representation of Figure 3 may visually amplify small shifts, the actual variation range for both pH and titratable acidity was extremely narrow in both treatments.

As shown in Figure 3A, the initial pH value of fresh-cut lettuce was 6.44, and only minor changes were observed over the 14-day storage period, which is in agreement with the study carried out by Kang et al. [27]. The increase in pH over time was significant only for UC samples, whereas the pH of EC samples did not change significantly.

Similarly, only small variations in TA were detected during storage (Figure 3B). The initial TA of 0.08% (w/w expressed as citric acid), consistent with previously reported values [18], showed a small and non-significant increase, remaining below 0.15% at day 8 and essentially stable thereafter. This behaviour is consistent with the literature descriptions of fresh-cut lettuce and other minimally processed vegetables, in which moderate metabolic activity may lead to the formation of weak organic acids and low-dissociation nitrogenous by-products. Although these compounds were not quantified in the present study, they have been reported to contribute to slight increases in titratable acidity while exerting minimal effects on pH due to their partial dissociation at the typical pH of lettuce tissues [25,28]. Overall, pH and titratable acidity were not suitable indicators for assessing the coating's effect under the conditions of this study.

3.3. Visual Appearance and Perception of Coated (EC) and Uncoated (UC) Salad

The visual appearance of ready-to-eat products, in general, and particularly of fresh-cut iceberg lettuce, plays a crucial role in the consumer's acceptance [29]. *Lactuca sativa* salad is valued for its crisp texture and pale green colour, both of which are easily affected by processing and storage conditions. However, iceberg lettuce remains highly sensitive to handling and environmental stress, and once it has been processed and packaged, its integrity can rapidly deteriorate. Consequently, its preservation continues to represent a major challenge in the fresh-cut food industry. The effect of the alginate-based emulsion coating application on the visual aspect of fresh-cut iceberg during storage at 4 °C is reported in Figure 4.

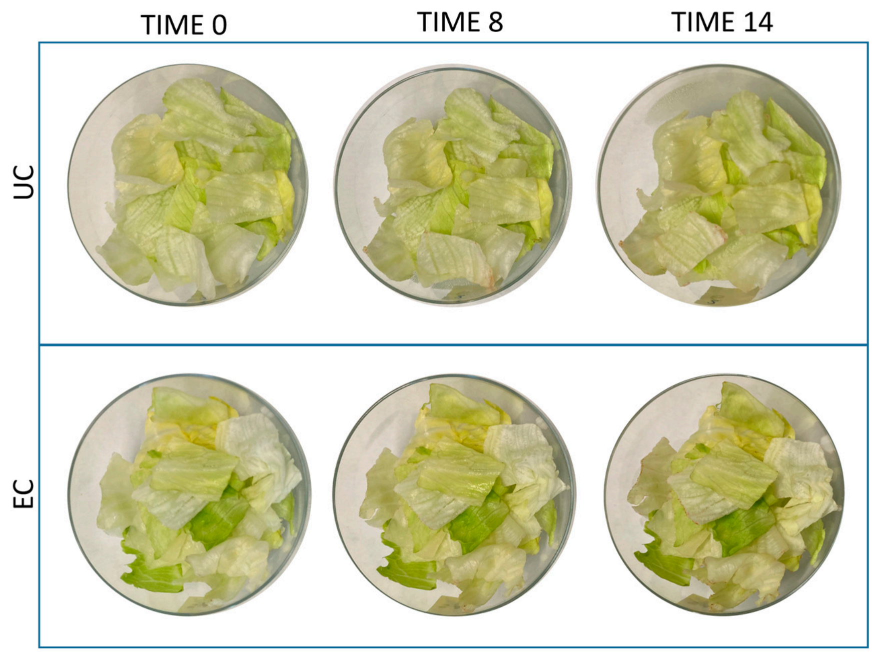

Figure 4. Visual appearance of coated salad (EC) and uncoated salad (UC) during the storage at 4 °C.

Commercially, packaged fresh-cut lettuce (e.g., PET clamshells or MAP bags) typically maintains an acceptable appearance for approximately 5–7 days before browning and tissue breakdown become more evident, as reported in previous studies [7,25]. Although a comparison with commercially packaged salads was not undertaken in this study, both EC and UC samples maintained acceptable visual quality over a similar time frame (Figure 4). Coated leaves showed visibly reduced edge browning for at least the first 8 days of storage, and overall acceptable appearance up to 14 days, whereas UC samples exhibited more evident browning and tissue degradation, particularly at the cut surfaces.

These findings suggest that the alginate-based emulsion coating may represent a suitable strategy, as it does not affect the appearance of the freshly cut iceberg lettuce and appears particularly effective during the first 8 days of storage. Comparable results have been reported by other authors, who observed that polysaccharide-based edible coatings helped to maintain quality, inhibit enzymatic browning, and delay senescence in fresh-cut lettuce [13]. Future studies should include a direct comparison with salads packaged in commercially available materials to better contextualize the performance of the coating under real distribution and retail conditions.

The influence of coating application on total colour changes (ΔE) of fresh-cut lettuce was investigated. Measurements of colour parameters were taken separately for white edges and green leaf areas (Table S1), and the ΔE calculated values are illustrated in Figure 5.

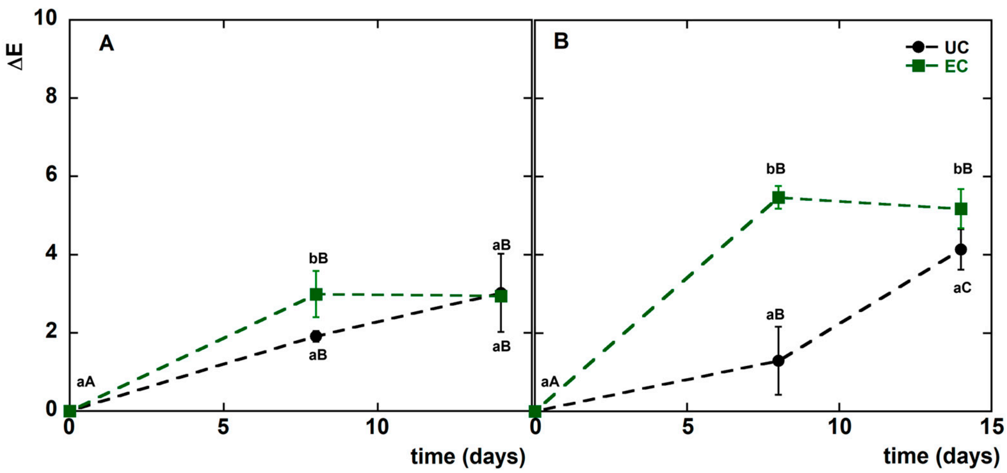

Figure 5. ΔE values variations during refrigerated storage of coated (EC) and uncoated (UC) salad during the storage at 4 °C: (**A**) white edges, (**B**) green leaves. Means with the same letters (lowercase: under different treatments for the same time; uppercase: under same treatment during different storage times) are not significantly different according to Duncan's test ($p > 0.05$). Vertical bars indicate standard deviation.

As shown in Figure 5, colour changes (ΔE) during storage were generally limited for both coated (EC) and uncoated (UC) lettuce. For the white edges (Figure 5A), ΔE values remained low and did not differ significantly between treatments ($p > 0.05$), indicating that the coating did not influence browning in these tissue regions.

For the green leaf portions (Figure 5B), higher ΔE values were observed in EC samples during the first 8 days, after which the values tended to stabilize and converge with those of UC samples by day 14.

Although EC samples exhibited a slightly higher early ΔE, this apparent discrepancy with the visual inspection (Figure 4) requires clarification. ΔE captures overall changes in surface optical properties and does not distinguish between true browning and variations in gloss or light scattering. The thin alginate layer could have transiently modified reflectance. This divergence highlights that the alginate coating did not negatively affect colour stability and may contribute to better preservation of the visual quality of iceberg lettuce under the storage conditions tested.

Remarkably, regardless of the whole effects related to the main chemical, physical, and chromatic parameters, it is also important to take into consideration the final consumer's acceptability of the commercialized ready-to-eat product. For this purpose, a triangle test was conducted to assess whether a group of untrained panellists could distinguish coated *Lactuca sativa* samples from uncoated ones. The panel consisted of 18 untrained assessors, a number that meets the minimum requirements commonly recommended for triangle tests in sensory analysis [24]. The panellists were explicitly asked to identify the coated sample within each triad, to evaluate the perceptibility of the edible coating. Given the limited number of assessors, the discrimination power of the triangle test should be interpreted as preliminary. The results of this investigation are summarized in Table 2.

Table 2. Triangle test data: total and correct number of responses.

Days	Total Number of Responses	Number of Correct Responses	Significance
0	36	24	$p < 0.05$
2	36	22	$p < 0.05$
7	36	16	$p > 0.05$
10	36	12	$p > 0.05$

As can be seen at time 0, immediately after the preparation of EC and UC, and after 2 days, the panel correctly identified the different samples, while at day 7, the number of correct answers (16) were not statistically significant ($p > 0.05$), meaning that the panellists were unable to distinguish between the uncoated and coated salad. This indication may be related to sensory characteristics, more perceptible immediately after coating application, which tend to diminish over time, as well as for the smell of the EO and the glossiness of the coating, as observed by other authors [30]. In fact, in the section dedicated to expressing the differences found between the samples, the terms smell and glossiness appear at time 0 and after 2 days, while the discriminants were mainly related to texture and visual appearance on the final days (7 and 10), though these were perceived correctly by only a limited number of judges. In the same test, the participants were asked to indicate whether they would purchase the product. The responses were consistently positive, with values of around 80% or higher (time 0 = 83%, time 2 days = 95%, time 7 days = 78%, time 10 days = 86%). The results obtained are encouraging, as the application of an edible coating does not always have a positive impact on the sensory characteristics of ready-to-use products. For example, an edible chitosan coating negatively affected the aroma of freshly cut lettuce [18], and some EOs, such as basil and thyme, were considered unacceptable when applied to lettuce [31].

Descriptive tests require trained panellists to generate detailed sensory profiles, whereas in this study, the key objective was to determine whether typical consumers could perceive differences between coated and uncoated samples. A consumer-based approach was, therefore, more appropriate for assessing product acceptability and market relevance.

The data from the consumer rating test were analyzed using a three-way ANOVA that evidenced the significant differences for samples relative to the different attributes. The results of the statistical analysis, expressed as the output of Fisher's least significant difference (LSD) test, are shown in Table 3. As reported, the colour was consistently perceived for both the samples and a small decrease in appreciation was assigned in terms of scores during storage. The odour was initially perceived as different in EC samples; nevertheless, the difference was less perceived over time. Crispiness was perceived by consumers as slightly lower for EC only for the second day of storage, while before and after storage, the two samples were equally scored.

Table 3. Consumer rating test for the coated (EC) and uncoated (UC) salad. Different letters in the same row indicate significant differences among samples (Fisher's LSD $p < 0.05$).

	Descriptors					
Time (Days)	**Colour**		**Odour**		**Consistency**	
	UC	**EC**	**UC**	**EC**	**UC**	**EC**
0	4.50 ^{a}	4.60 ^{a}	4.65 ^{a}	3.95 ^{b}	4.45 ^{a}	4.40 ^{a}
2	4.40 ^{a}	4.35 ^{a}	4.25 ^{a}	4.15 ^{a}	4.40 ^{a}	3.90 ^{b}
7	3.85 ^{a}	4.00 ^{a}	4.00 ^{a}	3.60 ^{b}	4.00 ^{a}	3.85 ^{a}
10	3.95 ^{a}	4.00 ^{a}	3.95 ^{a}	4.05 ^{a}	3.90 ^{a}	3.75 ^{a}

3.4. Microbial Analysis of Coated (EC) and Uncoated (UC) Salad

Because of its nutritional composition and high water content, fresh-cut lettuce provides an ideal environment for microbial growth. Furthermore, the different pre- and post-harvest operations make lettuce more susceptible to microbial contamination, which leads to quality deterioration and a consequent reduction in shelf life [32]. Psychrotrophic bacteria, like *Pseudomonas*, can grow at refrigeration temperatures and produce enzymes which degrade plant tissue causing changes in texture, flavour, and overall quality. Here, the effectiveness of the EO-enriched alginate coating in limiting microbial proliferation on minimally processed lettuce stored at 4 °C was assessed. The resulting counts of mesophilic (incubated at 30 °C) and psychrophilic (20 °C) bacteria expressed as log colony-forming units (log CFUs) per gram are shown in Figure 6.

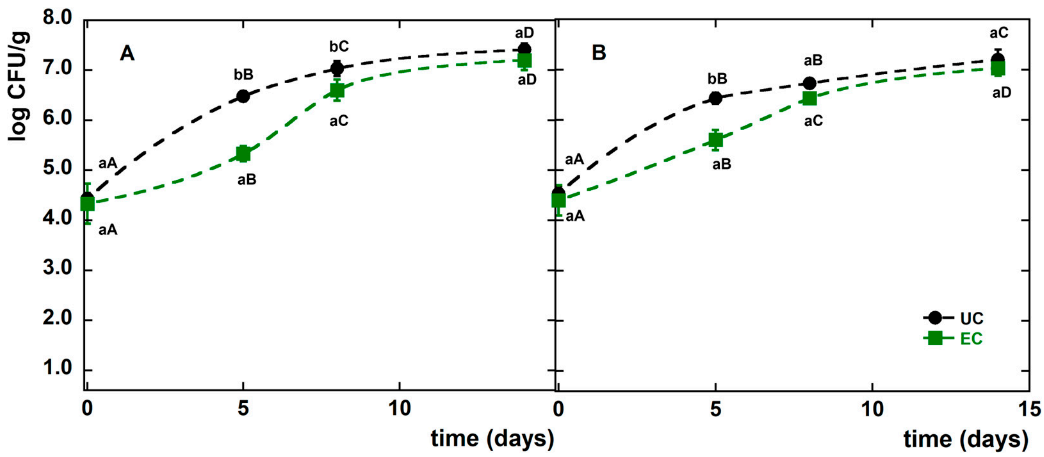

Figure 6. Evolution of (**A**) psychotropic and (**B**) mesophilic bacteria (log CFU/g) in fresh-cut lettuce stored at 4 °C for 14 days in uncoated (UC) and alginate-coated samples (EC). Means with the same letters (lowercase: under different treatments for the same time; uppercase: under the same treatment during different storage times) are not significantly different according to Tukey's test ($p > 0.05$). Vertical bars indicate standard deviations.

Figure 6A shows the evolution of psychrotrophic bacteria (log CFU/g) in fresh-cut lettuce stored at 4 °C for 14 days. On day 0, the EC sample exhibited an initial bacterial concentration (4.33 log CFU/g) comparable to that of the UC sample (4.43 log CFU/g). Although psychrotrophic counts increased over time in both tests, the coated sample consistently showed lower values. By day 8, psychrotrophic bacteria had reached a concentration of 6.60 log CFU/g in the EC sample and 7.03 log CFU/g in the UC sample. After 14 days of storage, the counts rose to 7.20 log CFU/g and 7.41 log CFU/g in the EC and UC samples, respectively, reflecting a gradual reduction in the inhibitory effect of the alginate-coating.

Figure 6B shows the dynamics of mesophilic bacteria under the same storage conditions. At day 0, the mesophilic counts were similar in EC (4.40 log CFU/g) and in UC samples (4.53 log CFU/g). A progressive increase in mesophilic counts was observed in both treatments over time, although the EC sample consistently exhibited slightly lower counts up to day 8, when levels reached 6.43 log CFU/g in EC and 6.73 log CFU/g in UC By day 14, mesophilic concentrations were very similar in the EC (7.03 log CFU/g) and UC (7.20 log CFU/g again suggesting a progressive loss of antimicrobial activity of the emulsion coating.

Yeasts, moulds, and lactic acid bacteria (LAB) remained below the detection limit in both uncoated and coated samples throughout storage. A reduction in psychrotrophic and mesophilic growth has also been reported for minimally processed lettuce coated with chitosan [22], marjoram EO [18], and encapsulated thyme oil, whereas the same EO in the free form oil did not effectively reduce the microbial load in lettuce [23].

Although the specific batch of EO used in this study was not chemically characterized, the antimicrobial activity observed is consistent with the behaviour typically reported for *Cymbopogon nardus* EO, in which compounds such as citronellal, geranial, and neral are generally identified as the main bioactive constituents [33].

4. Discussion

Bringing together the findings of the present study, which sought to extend the shelf life of fresh-cut products, it can be concluded that an effective way to enhance product longevity lies in the combined optimization of biocompatible surface treatments and storage conditions. Extending the expiry date, which as noted above is determined by the food business operators, is crucial for reducing food waste, since many products are discarded or remain unsold once they reach the declared "use-by" date.

To simulate an industrial application, 1% of alginate coating was selected based on the rheological characterization and applied onto the surface of the freshly cut lettuce; this reflects an easily scalable and reproducible process, thanks to the nebulizers already used in ready-to-eat salad production plants. Furthermore, spraying requires less coating than dipping and avoids excessive uptake, prolonged drying times, and thicker layers, all of which may adversely affect texture and appearance. In addition, repeated use of the same dipping bath can lead to dilution of the formulation and microbial contamination, rendering dipping less suitable for industrial-scale operations [34].

Ready-to-eat salads are perceived as fresh, safe, and healthy products and their popularity is closely linked to convenience, time saving, and the perception of high-quality meals. To preserve this perception, the coating formulation proposed here contained only three ingredients and no additional agents. Discrepancies in coating effects reported in the literature may be attributed to differences in application methods or to the use of crosslinking agents in other studies [12]. The results of the sensory analysis indicate that the alginate-based coating did not negatively influence key sensory attributes that drive consumer choice, showing favourable scores for odour, colour, and texture. The use of essential oil at a low concentration prevented any detrimental impact on the sensory characteristics of lettuce. Other authors have reported that the overall quality of fresh-cut iceberg lettuce deteriorates progressively during storage and becomes unacceptable after approximately 8 days [7].

The present results confirm the beneficial effect of the coating treatment, which appears to preserve product quality by slowing down microbial activity and metabolic changes observed in the first 8 days of storage. As reported by Jacxsens and coworkers [35], an increase in pH is typical of vegetables in which Gram-negative microorganisms play a major role in the spoilage, as observed here in the first 5 days of storage in the UC samples, where the microbial growth and pH increase were both greater than in the EC samples. The absence of lactic acid bacteria is consistent with good hygienic conditions maintained during processing and with the absence of anaerobic conditions throughout storage, in agreement with previous studies on fresh-cut lettuce [36].

These outcomes must also be considered in relation to the characteristics and dimensions of the packaging. Correct storage of fresh-cut salads (in terms of temperature, oxygen availability, use of additives, and so on) remains essential, as inadequate conditions can favour the growth of bacteria capable of proliferating at low temperatures. Although it is not possible to eliminate microbiological risks in ready-to-eat salads, as contamination may occur at several stages along the production chain, the proposed alginate-based edible coating appears to reduce and better control this risk before the product reaches consumers. Moreover, the findings suggest that treatment with the emulsion coating may have limited the enzymatic degradative activity of bacteria on plant tissues, contributing to the superior visual appearance of coated samples (EC) compared with untreated lettuce (UC).

From a shelf-life perspective, a comprehensive assessment of the effectiveness of edible coatings applied to fresh-cut salads will require further investigation. In particular, it will be essential to integrate data on the coating, used here as a primary protective layer, with information on the final packaging strategies adopted during commercialization, considering recent European packaging regulations and evaluating the protective effect against the principal pathogenic microorganisms.

5. Conclusions

An emulsion based on lemongrass essential oil in the presence of alginate was used as an edible coating to slow down spoilage of fresh-cut iceberg salad (*Lactuca sativa*). The formulation was effective in extending the post-harvest shelf life of salad by reducing its rate of deterioration.

The coating did not affect pH and titratable acidity, but it maintained the appearance and colour of the salad, with no visible browning of leaves or cut edges. During the first 8 days, the alginate-based coating effectively slowed the growth of psychrotrophic and mesophilic bacteria, thereby improving the microbial stability of fresh-cut lettuce stored at 4 °C. Moreover, the application of coating formulation did not negatively influence the sensory attributes of the coated salad compared with the control.

Overall, these findings demonstrate the potential of emulsion-based edible coatings as an alternative strategy for the fresh-cut salad industry. By helping to maintain product freshness, such systems could contribute to reducing food waste and thereby generate both economic and environmental benefits.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/colloids9060087/s1, Table S1: Colour coordinates (L*, a*, b*) values measured on the uncoated (UC) and coated salad (EC) during the storage at 4 °C.

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